Reporting Practice

Executive Summary

A security assessment was conducted on the target web application to identify potential vulnerabilities that could impact system security. The testing process included vulnerability scanning, manual testing, and exploitation attempts. The objective of the assessment was to evaluate the overall security posture of the system and identify weaknesses that could be exploited by attackers.

The assessment identified several security issues, including **SQL Injection** and **Weak Password Policies**. These vulnerabilities could allow attackers to access sensitive data or gain unauthorized access to user accounts. Immediate remediation actions are recommended to improve the security of the system and reduce the risk of exploitation.

Technical Findings

During the penetration test, multiple vulnerabilities were identified within the application**.**

SQL Injection:
The application does not properly validate user input in the login form. An attacker could inject malicious SQL queries to bypass authentication or access sensitive database information.

Weak Password Policy:
The system allows users to create weak passwords without enforcing complexity requirements. This increases the risk of brute-force attacks and unauthorized account access.

Remediation Plan

To improve the security of the system, the following actions should be implemented:

- Implement **input validation and prepared statements** to prevent SQL injection attacks.
- Enforce **strong password policies** including minimum length, special characters, and multi-factor authentication.
- Conduct regular **security testing and code reviews** to identify vulnerabilities early.
- Apply **security patches and updates** to all systems and applications.

Findings Table

Finding ID	Vulnerability	CVSS Score	Remediation
F001	SQL Injection	9.1	Implement input validation and prepared statements
F002	Weak Password	7.5	Enforce strong password complexity and MFA

Table 1: Findings Table

Network Attack Path Diagram

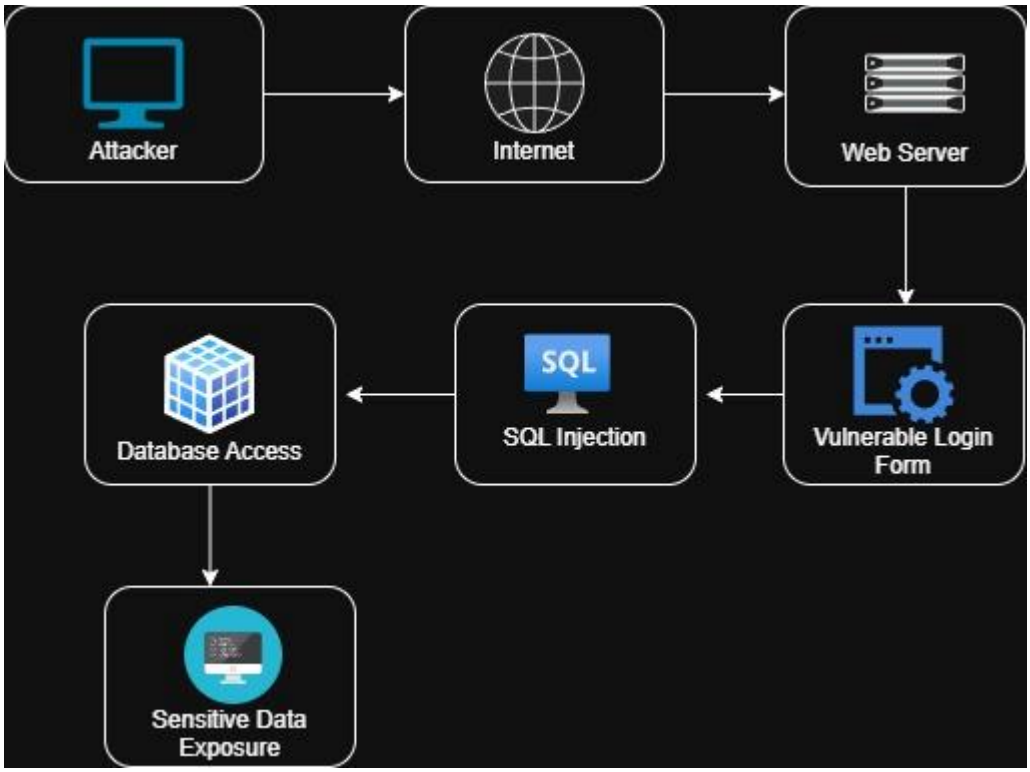

Figure 1: Network Attack Path Diagram

Non-Technical Summary for Managers

A security assessment was performed to evaluate the safety of the company's web application. The testing identified a few vulnerabilities that could potentially allow attackers to access sensitive information or compromise user accounts. The most critical issue discovered was a weakness that could allow unauthorized database access, while another issue involves weak password requirements that make accounts easier to guess. Although the system is generally functional, these issues could pose serious security risks if not addressed. Implementing stronger password policies and improving input validation in the application will significantly improve the overall security posture and help protect sensitive business and customer data.

Lab conclusion

This lab provided practical experience in creating a professional penetration testing report. The assessment identified critical vulnerabilities such as SQL Injection and weak password policies that could allow attackers to gain unauthorized access to sensitive information. By documenting the findings, assigning risk scores, and recommending remediation steps, the report demonstrates how security issues can be clearly communicated to both technical teams and management. Overall, the lab helped develop essential reporting skills required in penetration testing, including vulnerability documentation, risk analysis, and presenting security recommendations to improve the system's overall security posture.